• INDUSTRIAL DECISION INTELLIGENCE TWIN

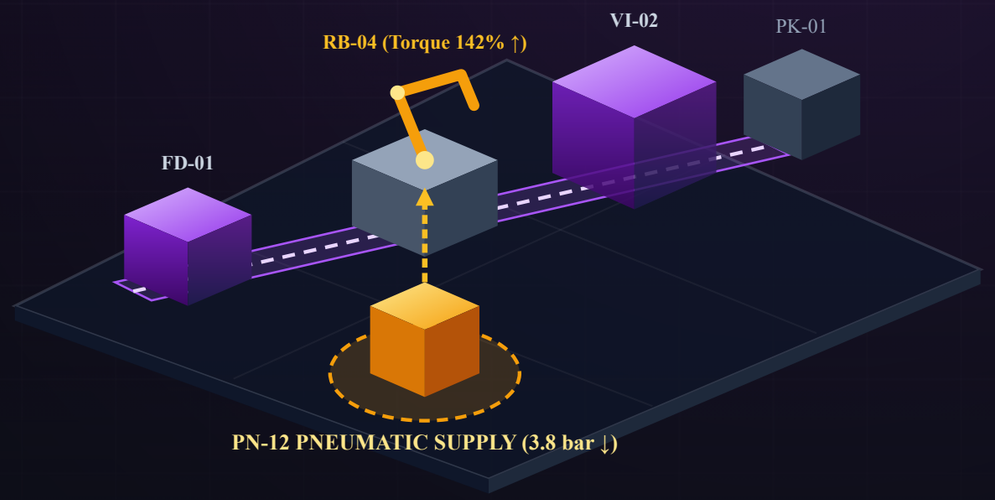
Don't predict the breakdown. Change the outcome.

Challenge naive downstream alarms. Respect physical spare-part inventory & safety constraints. Keep the plant supervisor in control.

0.624 bits	€11,439	-85%
Active Information Gain	Saved per Incident	Unplanned Downtime

FACTORY DIGITAL TWIN // ASSEMBLY LINE 01

▲ CASCADE ALERT: PN-12 → RB-04



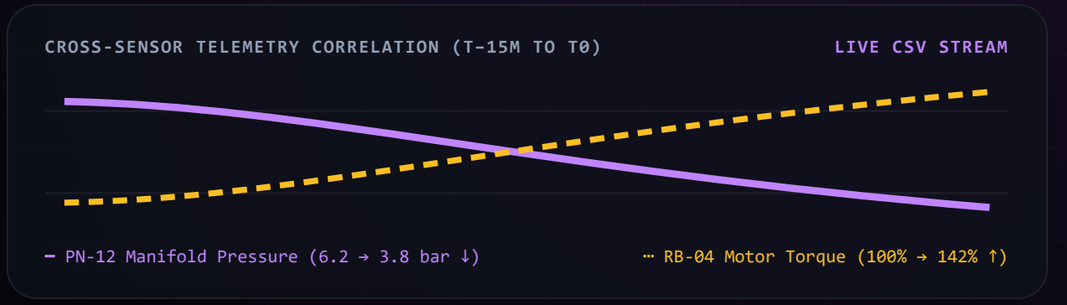
✦ ACTIVE DIAGNOSTIC DIRECTIVE

Do not replace RB-04 servo motor. Run 5-min leak verification on PN-12 first.

IG: 0.624 BITS

FOR THE LINE SUPERVISOR UNDER PRODUCTION PRESSURE

The alarm is not the explanation.



01 // UPSTREAM ROOT CAUSE (STATION PN-12)

NEXUSFORGE PATH

€18.50

30 min downtime

Pneumatic Viton O-Ring Seal Kit

A degraded upstream seal drops line pressure to `3.8 bar`, starving the downstream robotic actuator and forcing its drive motor to spike in torque. Verified via a 5-minute independent soap/ultrasonic check.

02 // DOWNSTREAM SYMPTOM ALARM (STATION RB-04)

NAIVE THRESHOLD ALERT

€2,450

210 min downtime

Unnecessary Servo Drive Teardown

Standard predictive tools react to the loudest downstream alarm (`RB-04 Torque Overload`). Ordering a servo swap burns 3.5 hours of production (€13,036 total loss) while leaving the air leak active.

ACTIVE DIAGNOSIS // REDUCE UNCERTAINTY BEFORE SPENDING

One observation. A different decision.

01 // PRIOR BELIEF STATE (HIGH ENTROPY: H = 1.58 BITS)

ACTION BLOCKED

Pneumatic seal leakage (PN-12)	51%
Manifold regulator instability	33%
Pressure transducer calibration drift	13%
Independent servo drive fault (RB-04)	03%

⟹

02 // POSTERIOR STATE AFTER OPERATOR TEST (H = 0.49 BITS)

REPAIR UNLOCKED

Pneumatic seal leakage (PN-12)	91%
Manifold regulator instability	08%
Pressure transducer calibration drift	01%
Independent servo drive fault (RB-04)	00%

+0.624 bits

Expected Information Gain: When two hypotheses compete (51% vs 33%), NexusForge refuses to guess. It computes the exact physical inspection that maximizes entropy reduction per minute.

$$IG(C; Y) = H(C) - E[H(C \mid Y)]$$

NOT A SCRIPTED ANSWER // A DECISION ENGINE YOU CAN CHALLENGE

Change the constraints. Watch the plan adapt.

01

UNVERIFIED SIGNAL

Inspect First

Holds spare-part commitment until the 5-minute discriminating leak check is confirmed by an operator.

02

CONFIRMED + STOCK > 0

Authorize Fix

Generates Work Order for `PN-12` Viton seal kit when inventory (`15 units`) fits the 90m window.

03

STOCK = 0 (LEAD: 180m)

Block Teardown

Right diagnosis ≠ valid action. Prevents stripping a line when supplier delivery exceeds the 90m deadline.

04

CABINET TEMP > LIMIT

Hard Escalate

Deterministic thermal safety rules override cost optimization and enforce immediate line isolation.

€11,439

Direct Net Savings per Avoided False-Alarm Teardown

Compares **€13,036** unverified reactive motor swap (210m lost production + €2,450 hardware) against **€1,597** verified seal replacement (30m downtime + €18.50 part).

−180 MIN DOWNTIME | ZERO WASTE

OPEN MODELS. DETERMINISTIC CONSTRAINTS. A HUMAN SIGNATURE.

Intelligence proposes. Policy constrains.

01 INPUT Sensor CSV + OEM Manuals → 02 EVOLUS RAG Extractors & Knowledge Library → 03 AMD ROCM Open-Weight vLLM + Solver → 04 EVOLUS ACTION Signed JSON Workflow Dispatch

BUILDING BLOCKS // INGESTION — EVOLUS API

Document Extractors & Libraries

Parses maintenance manuals and telemetry streams (`pressure_bar`, `torque_pct`, `temp_c`) via **Evolus Knowledge Libraries** and structured JSON **Document Extractors**, binding every hypothesis to exact source quotes.

INFERENCE ENGINE // PRIVACY — AMD CLOUD

Open Models on AMD ROCm

Executes agentic reasoning on open-weight models served via **vLLM on AMD Instinct GPUs (ROCm)** connected to the Evolus agent endpoint—delivering 100% air-gapped factory IP protection and predictable token economics.

EXECUTION // GOVERNANCE — WORKFLOWS

Human Sign-Off & Evolus Dispatch

Zero autonomous plant writes. Once the supervisor signs the checklist, NexusForge calls `POST /api/v1/workflows/{id}/execute` to dispatch auditable Work Orders (`NF-3A65A7CA`) to ERP & procurement.

52/52 **Unit Tests & 44/44 Decision Contracts Passed**
Dockerized Node.js runtime · Tamper-evident SQLite audit chain · MIT Licensed

AMD ROCm vLLM + Evolus Workflows: LIVE & VERIFIED

• FROM A SLIDE DECK TO A VERIFIABLE DECISION

Your evidence. Your constraints. Your next move.

Don't trust a static presentation. Challenge the live engine right now with contradictory sensor logs, zero spare-parts stock, or thermal safety breaches.

✦ Switch to Live NexusForge Workspace

LIVE DEMO PROTOCOL // 3 INTERACTIVE STEPS

REAL-TIME API

01

Inspect the 3D Cascade & Evidence Studio

Observe the `PN-12` pressure drop starving `RB-04`, or load any of the 8 adversarial telemetry & inventory scenarios.

02

Stress-Test the Counterfactual Lab

Confirm the `0.624 bits` leak test to jump confidence from 51% to 91%—then set `Stock = 0` to watch the solver block premature teardown.

03

Sign & Trigger Live Evolus Automation

Approve the work order in Action Center under `Paolo Chignoli` and verify the real-time execution inside the Evolus Automations dashboard.